ChE 210 Homework Set #5 Due Friday, 27 February 2026

(1) Murphy 3.21

(2) Murphy 3.29

(3) Murphy 3.41 (To find h_f, the final height of the fluid in the tank, you will need to use Excel Solver to solve the nonlinear equation that arises. On page 676 of Murphy, the solution for this problem is given as 5.1 kg of X. A better value is 5.394 kg of X.)

(4) Phosphoric acid (PA) can be produced by reacting calcium phosphate (CP) with a stoichiometric quantity of sulfuric acid (SA). Calcium sulfate (CS) is also a byproduct of this reaction. Two streams enter the reactor: (**Stream 1**) has a species mass flow rate 10,000 kgCP/hr of CP which is suspended in an unknown flow rate of water (W) and (**Stream 2**) a 95 wt% aqueous solution of SA. There are two exit streams: (**Stream 3**) a 40 wt% aqueous solution of PA and (**Stream 4**) a solid of CS and water with a mole ratio of 2 mole of water for each mole of CS, that is, $CaSO_4 \cdot 2H_2O$, which is commonly call gypsum.

(a) Assuming steady operation, solve for all of the unknown flow rates. Show that the mass flow rate into the process equals the mass flow rate out of the process.

(b) If the operation is started when the tank contains 10,000 kg of 20 wt% phosphoric acid aqueous solution, what is the concentration in the tank after one hour?

(5) A tank containing 1000 L of 95 wt% ethanol solution in water operates at 20°C and steady state with a continuous flow in and out of 100 L/min of 95 wt% ethanol. At t = 0, the input ethanol solution flow is suddenly stop and immediately replaced by a flow of 100 L/min of pure water entering the tank. The pure water flowing in is now diluting the ethanol solution in the tank. Assume the total mass of the solution in the tank remains constant. Assume that the tank is well stirred throughout the process.

DATA: Use the webpage:
https://www.handymath.com/cgi-bin/ethanolwater3.cgi?submit=Entry
for the density of an ethanol / water mixture in kg/L at 20℃. This data is also posted on Blackboard under documents.

(a) How long in minutes will it take for the mass fraction of ethanol in the tank to fall from 95 wt% to 5 wt%.

(b) Recalculate the time for the case when the total volume of the solution in the tank remains constant rather than the total mass. Again dilute the ethanol from 95 wt% to 5 wt%. Give the dilution time in minutes.

Hint: Write mass balances on the total mass and the ethanol mass. Decide what the two unknown variables are. The data gives you the mixture density as a function of mass fraction or vice versa. Combine the two mass balances to eliminate one of the unknowns to create one differential equation with one unknown. You will want to simplify this equation and then solve it.

P3.21 A 12-oz. mug of coffee contains 200 mg of caffeine. Caffeine is eliminated from the body at a rate of $dm_{c,sys}/dt = -0.116 m_{c,sys}$, where $m_{c,sys}$ is the mass of caffeine in the body, $m_{c,sys}$ is in mg and t is in h. After you chug down a mug of coffee, how long will it take for the caffeine in your body to drop to 100 mg? You drink one 12-oz. mug at 6 A.M. and another at 2 P.M. Plot the caffeine content of your body as a function of time for one 24-h interval. If you are having trouble falling asleep at 11 P.M., would it help much to cut out that second mug?

$\frac{dm}{dt} = -0.116m \qquad m(t) = m_0 e^{-0.116t}$

a) $100 = 200e^{-0.116t}, \quad 0.5 = e^{-0.116t}$

$t = \frac{\ln 2}{0.116} = 5.975\,h$ ✓

b) $0 \le t \le 8, \quad m(t) = 200e^{-0.116t}$

$8 \le t \le 24, \quad m(t) = 200e^{-0.116t} + 200e^{-0.116(t-8)}$

6am $(t=0)$: 200 mg

12pm $(t=6)$: $200e^{-0.696} = 99.72$ mg

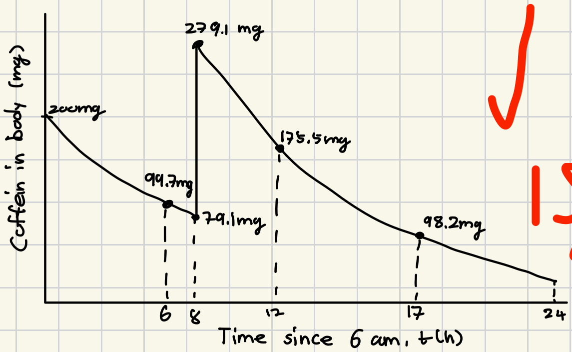
before 2pm $(t=8^-)$: $200e^{-0.928} = 79.07$ mg

after 2pm $(t=8^+)$: $79.07 + 200 = 279.07$ mg

6pm $(t=12)$: $200e^{-1.392} + 200e^{-0.464} = 175.47$ mg

11pm $(t=17)$: $200e^{-1.972} + 200e^{-1.044} = 98.24$ mg

next 6am $(t=24)$: $200e^{-2.784} + 200e^{-1.856} = 43.62$ mg



✓

15/15

c) 11pm, $t=17$

With 2pm mug: $m(17) = 200e^{-0.116\times17} + 200e^{-0.116\times9} = 98.2$ mg ✓

without 2pm mug: $m(17) = 200e^{-0.116\times17} = 27.8$ mg

difference ≈ 70.4 mg → Yes, cutting the second mug helps.

P3.29 Immunotoxins are molecules designed to specifically kill cancer cells without damaging healthy noncancerous cells. Immunotoxins must accumulate inside cancer cells before they can be effective. A start-up biotech company has discovered a new immunotoxin. The company is trying to raise venture capital. You are trying to decide if you should invest your millions. The start-up company provides the following data. The immunotoxin enters the typical cancer cell at a rate of 62,000 molecules per hour. Some of the immunotoxin is degraded inside the cancer cell by enzymes. The rate of degradation is estimated to be $2700(1 - e^{-0.3t})$ molecules degraded per hour, where t is time in hours since the cancer cell was first exposed to the immunotoxin. The cancer cell also spits out some of the immunotoxin; the rate of immunotoxin leaving the cancer cell is estimated to be 57,000 molecules per hour. You find independent scientific literature that indicates that, for an immunotoxin to successfully kill a cancer cell, there must be an accumulation of at least 30,000 immunotoxin molecules inside the cell after 8 hours. Should you invest?

$2700(1-e^{-0.3t})$ molecules/h

Balance

$\frac{dN}{dt} = 62000 - 57000 - 2700(1-e^{-0.3t})$

$= 2300 + 2700e^{-0.3t}$

$N(t) = 2300t + 2700\int_0^t e^{-0.3t}dt = 2300t + 2700\left(\frac{1-e^{-0.3t}}{0.3}\right)$

$N(t) = 2300t + 9000(1-e^{-0.3t})$

$t=8, \quad N(8) = 2300\times8 + 9000(1-e^{-0.3\times8})$

$= 26583.5$ ✓

$\therefore N(8) \approx 2.66\times10^4 < 3.00\times10^4$ ✓

Requirement is at least 30000 molecules after 8 hours, hence, do not invest.

✓ 15/15

P3.41 As part of a start-up process, a liquid solution containing a very hazardous material (compound X) is pumped into an empty feed tank. The cylindrical feed tank is 1 m in diameter and 3 m high. The solution (density = 1000 kg/m³, 0.1 kg X/kg solution) is pumped in at a rate of 40 kg/min. At a point 1 m up the tank wall, a corroded spot gives way, and a leak develops. The leak rate gets worse as the tank fills, with the rate increasing as the square root of the height of the liquid above the leak:

$\dot{m}_{leak}$ (kg/min) = 4 × (liquid height in tank − tank height at leak point)$^{0.5}$

How much compound X has leaked out, when you walk by the tank 40 minutes after the start of the tank filling process and notice the leak?

$A = \pi 0.5^2 = 0.7854\,m^2$

$Q_{in} = 40\frac{kg}{min} \times \frac{1}{1000}\frac{m^3}{kg} = 0.04\,m^3/min$

$\dot{m}_{leak} = 4(h-1)^{1/2}$ kg/min

$Q_{leak} = \frac{\dot{m}_{leak}}{\rho} = 0.004(h-1)^{1/2}\,m^3/min$

a) befor leak $h \le 1$,

$A\frac{dh}{dt} = Q_{in}, \quad \frac{dh}{dt} = \frac{0.04}{0.7854} = 0.05093\,m/min$

$t_1 = \frac{1}{0.0593} = 19.635$ min

leakage occurs from $t = 19.635 \sim 40$ min

b) $A\frac{dh}{dt} = 0.04 - 0.004(h-1)^{1/2}$

$y = \sqrt{h-1}, \quad h = y^2+1, \quad dh = 2y\,dy$

$2Ay\frac{dy}{dt} = 0.04 - 0.004y$

$dt = \frac{2Ay}{0.04-0.004y}dy$

$\frac{2A}{0.004} = 500A = 392.699$

$dt = 392.699\frac{y}{10-y}dy$

$\int_{t_1}^{40} dt = 392.699\int_0^{y_{40}} \frac{y}{10-y}dy$

$40 - t_1 = 392.699\left(-y_{40} + 10\ln\left(\frac{10}{10-y_{40}}\right)\right)$

$h=1, \quad t_1 = \frac{1}{0.04/A} = \frac{A}{0.04} = 19.63495\,min$

$40 - t_1 = 20.37\,min$

$-y_{40} + 10\ln\left(\frac{10}{10-y_{40}}\right) = 0.052$

$y_{40} = 0.984$

$h(40) = y_{40}^2 + 1 = 1.969\,m$ ✓

c) $\dot{m}_{leak} = 4\sqrt{h-1}$ kg/min

$y = \sqrt{h-1}, \quad \dot{m}_{leak} = 4y$

$m_{leak,soln} = \int_{t_1}^{40} \dot{m}_{leak}dt = \int_{t_1}^{40} 4y\,dt$

$= \int_0^{y_{40}} 4y\left(392.699\frac{y}{10-y}dy\right) = 1570.796\int_0^{y_{40}}\frac{y^2}{10-y}dy$

$\frac{y^2}{y-10} = y + 10 + \frac{100}{y-10}$

$\int_0^{y_{40}}\frac{y^2}{10-y}dy = \left[-\frac{y^2}{2} - 10y - 100\ln|y-10|\right]_0^{y_{40}}$

$= -\frac{y_{40}^2}{2} - 10y_{40} + 100\ln\left(\frac{10}{10-y_{40}}\right)$

$m_{leak,soln} = 1570.796\left[-\frac{y_{40}^2}{2} - 10y_{40} + 100\ln\left(\frac{10}{10-y_{40}}\right)\right]$

$y_{40} = 0.98444149$

$-\frac{y_{40}^2}{2} = -0.48428$

$\therefore m_{leak,soln} = 53.94147\,kg$ ✓

0.1 kg X per kg solution

$m_{X,leak} = 0.1\,m_{leak,soln} = 0.1(53.94147)$

$\therefore$ Mass of X leaked $\approx 5.394\,kg$ ✓

20/20

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(b) If the operation is started when the tank contains 10,000 kg of 20 wt% phosphoric acid aqueous solution, what is the concentration in the tank after one hour?

Stream 1: $\dot{m}_{CP} = 10000$ kg/h, unknown water

Stream 2: 95 wt% SA, unknown flow

Stream 3: 40 wt% PA, unknown flow

Stream 4: $CaSO_4 \cdot 2H_2O$, unknown flow

$$Ca_3(PO_4)_2 + 3H_2SO_4 \rightarrow 2H_3PO_4 + 3CaSO_4$$

$Ca_3(PO_4)_2$: 310.174

H_2SO_4 : 98.077

H_3PO_4 : 97.994

$CaSO_4 \cdot 2H_2O$: 172.169

a) $\dot{n}_{cp} = \frac{10000}{310.174} = 32.24$ kmol/h

$\dot{n}_{SA} = 3\dot{n}_{cp} = 96.72$ kmol/h

$\dot{n}_{PA} = 2\dot{n}_{cp} = 64.48$ kmol/h

$\dot{n}_{CaSO_4} = 3\dot{n}_{cp} = 96.72$ kmol/h

St2: $\dot{m}_{SA} = \dot{n}_{SA}(98.077) = 9485.998$ kg/h

$\dot{m}_2 = \frac{\dot{m}_{SA}}{0.95} = 9985.261$ kg/h

$\dot{m}_{2,w} = 0.05\dot{m}_2 = 499.263$ kg/h

St3: $\dot{m}_{PA} = \dot{n}_{PA}(97.994) = 6318.647$ kg/h

$\dot{m}_3 = \frac{\dot{m}_{PA}}{0.4} = 15796.617$ kg/h

$\dot{m}_{3,w} = 0.6\dot{m}_3 = 9477.97$ kg/h

St4: $\dot{m}_4 = \dot{n}_{CaSO_4}(172.169) = 16652.169$ kg/h

$\dot{m}_{hyd} = \dot{n}_{CaSO_4}(2 \times 18.015) = 3483.233$ kg/h

Water in = Water out

$\dot{m}_{1,w} + \dot{m}_{2w} = \dot{m}_{3,w} + \dot{m}_{4w}$

$\dot{m}_{1,w} = 9477.97 + 3483.233 - 499.263 = 12463.526$ kg/h

$\dot{m}_{in} = 10000 + 12463.526 + 9985.261 = 32448.787$ kg/h

$\dot{m}_{out} = 15796.617 + 16652.169 = 32448.787$ kg/h

$\dot{m}_{in} = \dot{m}_{out}$

b) x = mass fraction of PA in tank

$\dot{m}_{PA,gen} = 6318.647$ kg/h

$\dot{m}_3 = 15796.617$ kg/h

PA balance: $\frac{d(Mx)}{dt} = \dot{m}_{PA,gen} - x\dot{m}_3$

M constant, $\frac{dx}{dt} = \frac{\dot{m}_{PA,gen}}{M} - \frac{\dot{m}_3}{M}x$

Steady value: $x_s = \frac{\dot{m}_{PA,gen}}{\dot{m}_3} = 0.4$

$$x = x_s + (x_0 - x_s)e^{-(\dot{m}_3/M)t}$$

$x_0 = 0.2$, $t = 1$h,

$x = 0.4 + (0.2 - 0.4)e^{-(15796.617/10000)}$

$= 0.35879 \Rightarrow 35.9\%$

$\therefore$ 35.9 wt% PA after 1 hour

20/20

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$V_0 = 1000$ L, $x_0 = 0.95$

$\rho_w = \rho(0\%) = 0.99823$ kg/L

$\rho(5\%) = 0.98938$ kg/L

$\rho(95\%) = 0.80424$ kg/L

a) $M = V_0\rho(95\%) = 804.24$ kg

$\dot{m}_{in} = 100(0.99823) = 99.823$ kg/min

$\dot{m}_{in} = \dot{m}_{out}$

x = ethanol mass fraction in tank

ethanol in = 0, ethanol out = $x\dot{m}_{out}$

$M\frac{dx}{dt} = -\dot{m}_{out}x$, $x = x_0 e^{-kt}$

$k = \frac{\dot{m}_{out}}{M} = \frac{99.823}{804.24} = 0.1241 \text{ min}^{-1}$

t for x: $0.95 \rightarrow 0.05$,

$0.05 = 0.95e^{-kt}$, $t = \frac{1}{k}\ln\left(\frac{0.95}{0.05}\right) = 23.72$ min

b) $V = 1000$ L

$Q_{in} = Q_{out} = 100$ L/min

Total tank mass: $M = \rho(x)V$

Ethanol mass: $m_E = x\rho(x)V$

Ethanol balance: $\frac{d}{dt}(x\rho(x)V) = -(x\rho(x))Q_{out}$

$\frac{d}{dt}(x\rho(x)) = -\frac{Q_{out}}{V}x\rho(x) = -0.1x\rho(x)$

$\frac{1}{u}du = -0.1dt$

$\int_{u(0)}^{u(t)} \frac{1}{u}du = \int_0^t -0.1dt$, $\ln\left(\frac{u(t)}{u(0)}\right) = -0.1t$

$u(t) = u(0)e^{-0.1t}$

$x\rho(x) = x_0\rho(x_0)e^{-0.1t}$

$x_f = 0.05$, $t = t_f$,

$0.05\rho(0.05) = 0.95\rho(0.95)e^{-0.1t_f}$

$e^{-0.1t_f} = \frac{0.05\rho(0.05)}{0.95\rho(0.95)}$

$t_f = 10\ln\left(\frac{0.95\rho(0.95)}{0.05\rho(0.05)}\right) = 27.37$ min

So the constant volume case takes longer than the constant mass case.

30/30